Name
Date
Course

Applications (Part II)

The Law of Uninhibited Growth/Decay

Any amount "A" that changes over time according to the function below is said to follow the law of uninhibited growth ($k>0$) or decay ($k<0$),

$$A(t)=A_0e^{kt}$$

where $A(t)$ is the amount of a substance after time "t" has passed, "A_0" is the initial amount, and "k" is an experimentally determined constant.

① The population of a colony of bacteria follows the law of uninhibited growth ($k=0.22$). If a colony starts with 82 bacteria, how many will be present after 19 days?

$$A(t)=A_0e^{kt}$$

$$A(19)=82e^{0.22(19)}$$

$$A(19)=82e^{4.18}$$

$$A(19)=82(65.3658\ldots)$$

$$A(19)\approx\boxed{5{,}360 \text{ bacteria}}$$

② Cobalt-60 converts to nickel according to the law of uninhibited decay ($k=-0.13$). If 65 grams of cobalt-60 is present in a sample, how much would be left after 10 years?

$$A(t)=A_0e^{kt}$$

$$A(10)=65e^{-0.13(10)}$$

$$A(10)\approx\boxed{17.7 \text{ grams cobalt-60}}$$

③ The population of a culture of cells follows the law of uninhibited growth ($k=0.32$). If 200 cells are initially present, what will the size of the culture be after 30 days?

$$A(t)=A_0e^{kt}$$

$$A(30)=200e^{0.32(30)}$$

$$A(30)\approx\boxed{2{,}952{,}956 \text{ cells}}$$

④ Strontium-90 converts to zirconium according to the law of uninhibited decay ($k=-0.024$). If 98 grams of strontium-90 is present, how many grams would be left after 8.6 years?

$$A(t)=A_0e^{kt}$$

$$A(8.6)=98e^{-0.024(8.6)}$$

$$A(8.6)\approx\boxed{79.7 \text{ grams strontium-90}}$$

⑤ A suspension of yeast grew from 622 organisms to 10,202 organisms in 35 minutes. If this same strain started with 1,000 organisms, how many would there be after 139 minutes?

I) Find K.

$$A(t) = A_0 e^{kt}$$

$$10{,}202 = 622 e^{k(35)}$$

$$\frac{10{,}202}{622} = \frac{622 e^{35k}}{622}$$

$$16.401\ldots = e^{35k}$$

$$35k = \log_e 16.401\ldots$$

$$35k = \ln 16.401\ldots$$

$$k = \frac{\ln 16.401\ldots}{35}$$

$$k = 0.0799\ldots$$

II) $A(t) = A_0 e^{kt}$

$$A(139) = 1{,}000 e^{(0.0799\ldots)(139)}$$

$$\boxed{A(139) \approx 66{,}814{,}151 \text{ organisms}}$$

⑥ In 3 years, a radioactive substance decreases from 56 grams to 22 grams. How much of this same radioactive material would be left if we started with an 8 gram sample and monitored it for a year?

I) Find k.

$$A(t) = A_0 e^{kt}$$

$$22 = 56 e^{k(3)}$$

$$\frac{22}{56} = \frac{56 e^{3k}}{56}$$

$$0.3928\ldots = e^{3k}$$

$$3k = \log_e 0.3928\ldots$$

$$3k = \ln 0.3928\ldots$$

$$k = \frac{\ln 0.3928\ldots}{3}$$

$$k = -0.311\ldots$$

II) $A(t) = A_0 e^{kt}$

$$A(1) = 8 e^{(-0.311\ldots)(1)}$$

$$\boxed{A(1) \approx 5.859 \text{ grams}}$$

7. A suspension of yeast grew from 766,000 organisms to 1,862,000 organisms in 13 minutes. If this same strain started with 123,000 organisms, how many would there be after 45 minutes?

I) Find k

$$A(t) = A_0 e^{kt}$$

$$1{,}862{,}000 = 766{,}000 e^{k(13)}$$

$$\frac{1{,}862{,}000}{766{,}000} = \frac{766{,}000 e^{13k}}{766{,}000}$$

$$2.43\ldots = e^{13k}$$

$$13k = \log_e 2.43\ldots$$

$$13k = \ln 2.43\ldots$$

$$k = \frac{\ln 2.43\ldots}{13}$$

$$k = 0.06832\ldots$$

II) $A(t) = A_0 e^{kt}$

$$A(45) = 123{,}000 e^{(0.06832\ldots)(45)}$$

$$\boxed{A(45) \approx 2{,}661{,}931 \text{ organisms}}$$

8. In 121 minutes a radioactive substance decreases from 4.84 grams to 2.2 grams. How much of this same radioactive material would be left if we started with a 9.44 gram sample and monitored it for 60 minutes?

I) Find k.

$$A(t) = A_0 e^{kt}$$

$$2.2 = 4.84 e^{k(121)}$$

$$\frac{2.2}{4.84} = \frac{4.84 e^{121k}}{4.84}$$

$$0.\overline{45} = e^{121k}$$

$$121k = \log_e 0.\overline{45}$$

$$121k = \ln 0.\overline{45}$$

$$k = \frac{\ln 0.\overline{45}}{121}$$

$$k = -0.00651$$

II) $A(t) = A_0 e^{kt}$

$$A(60) = 9.44 e^{(-0.00651\ldots)(60)}$$

$$\boxed{A(60) \approx 6.385 \text{ grams}}$$

Half Life: The time that it takes for a substance to lose 50% of itself.

The decay of radioactive material follows the law of uninhibited decay. Use this knowledge to answer the following questions.

⑨ The half life of cesium-137 is 30 years. If a scientist obtains 4 grams of cesium-137, how much will be left after 96 years?

I) Find K.

$$A(t) = A_0 e^{kt}$$

$$\frac{1}{2}A_0 = A_0 e^{k(30)}$$

$$\frac{\frac{1}{2}A_0}{A_0} = \frac{A_0 e^{30k}}{A_0}$$

$$\frac{1}{2} = e^{30k}$$

$$30k = \log_e\left(\frac{1}{2}\right)$$

$$30k = \ln\left(\frac{1}{2}\right)$$

$$K = \frac{\ln\left(\frac{1}{2}\right)}{30}$$

$$K = -0.0231\ldots$$

II) $A(t) = A_0 e^{kt}$

$$A(96) = 4e^{(-0.0231\ldots)(96)}$$

$$\boxed{A(96) \approx 0.435 \text{ grams cesium-137}}$$

⑩ The half-life of plutonium-238 is 87 years. If a physicist obtains 0.23 grams, how much will be left after 60 years?

I) Find K.

$$A(t) = A_0 e^{kt}$$

$$\frac{1}{2}A_0 = A_0 e^{k(87)}$$

$$\frac{\frac{1}{2}A_0}{A_0} = \frac{A_0 e^{k(87)}}{A_0}$$

$$\frac{1}{2} = e^{87k}$$

$$87k = \log_e\left(\frac{1}{2}\right)$$

$$87k = \ln\left(\frac{1}{2}\right)$$

$$K = \frac{\ln\left(\frac{1}{2}\right)}{87}$$

$$K = -0.00796\ldots$$

II) $A(t) = A_0 e^{kt}$

$$A(60) = 0.23e^{(-0.00796\ldots)(60)}$$

$$A(60) \approx \boxed{0.1426 \text{ grams plutonium-238}}$$

(11) The half life of iron-59 is 45 days. If a scientist obtains 15 grams of iron-59, how much will be left after 3 days?

I) Find k.

$A(t) = A_0 e^{kt}$

$\frac{1}{2}A_0 = A_0 e^{k(45)}$

$\frac{\frac{1}{2}A_0}{A_0} = \frac{A_0 e^{k(45)}}{A_0}$

$\frac{1}{2} = e^{45k}$

$45k = \log_e\left(\frac{1}{2}\right)$

$45k = \ln\left(\frac{1}{2}\right)$

$k = \frac{\ln\left(\frac{1}{2}\right)}{45}$

$k = -0.0154...$

II) $A(t) = A_0 e^{kt}$

$A(3) = 15e^{(-0.0154...)(3)}$

$A(3) \approx \boxed{14.32 \text{ grams iron-59}}$

(12) The half-life of iodine-131 is 8.1 days. If a doctor obtains 100 grams, how much will be left after 15 days?

I) Find k.

$A(t) = A_0 e^{kt}$

$\frac{1}{2}A_0 = A_0 e^{k(8.1)}$

$\frac{\frac{1}{2}A_0}{A_0} = \frac{A_0 e^{8.1k}}{A_0}$

$\frac{1}{2} = e^{8.1k}$

$8.1k = \log_e\left(\frac{1}{2}\right)$

$8.1k = \ln\left(\frac{1}{2}\right)$

$k = \frac{\ln\left(\frac{1}{2}\right)}{8.1}$

$k = -0.0855...$

II) $A(t) = A_0 e^{kt}$

$A(15) = 100e^{(-0.0855...)(15)}$

$\boxed{A(15) \approx 27.7 \text{ grams iodine-131}}$

⑬ The population of a culture of cells follows the law of uninhibited growth. If the population doubles in 3 hours, how long will it take for 2,000 cells to increase to one million cells?

I) Find k.

$A(t) = A_0 e^{kt}$

$2A_0 = A_0 e^{k(3)}$

$\frac{2A_0}{A_0} = \frac{A_0 e^{3k}}{A_0}$

$2 = e^{3k}$

$3k = \log_e 2$

$3k = \ln 2$

$k = \frac{\ln 2}{3}$

$k = 0.231\ldots$

II) $A(t) = A_0 e^{kt}$

$1{,}000{,}000 = 2{,}000 e^{(0.231\ldots)t}$

$\frac{1{,}000{,}000}{2000} = \frac{2{,}000 e^{(0.231\ldots)t}}{2000}$

$500 = e^{(0.231\ldots)t}$

$(0.231\ldots)t = \log_e 500$

$\frac{(0.231\ldots)t}{0.231\ldots} = \frac{\ln 500}{0.231\ldots}$

$t \approx \frac{\ln(500)}{\text{Ans}}$

$\boxed{t \approx 26.897 \text{ hours}}$

⑭ The population of a colony of bacteria follows the law of uninhibited growth. If the size of the colony triples in 2 days, how long will it take 48 bacteria to increase to 72 million bacteria?

I) Find k.

$A(t) = A_0 e^{kt}$

$3A_0 = A_0 e^{k(2)}$

$\frac{3A_0}{A_0} = \frac{A_0 e^{2k}}{A_0}$

$3 = e^{2k}$

$2k = \log_e 3$

$2k = \ln 3$

$k = \frac{\ln 3}{2}$

$k = 0.5493\ldots$

II) $A(t) = A_0 e^{kt}$

$72{,}000{,}000 = 48 e^{(0.5493\ldots)t}$

$\frac{72{,}000{,}000}{48} = \frac{48 e^{(0.5493\ldots)t}}{48}$

$1{,}500{,}000 = e^{(0.5493\ldots)t}$

$(0.5493\ldots)t = \log_e 1{,}500{,}000$

$(0.5493\ldots)t = \ln 1{,}500{,}000$

$t = \frac{\ln 1{,}500{,}000}{0.5493\ldots}$

$\boxed{t \approx 25.89 \text{ days}}$

Review on Finding the Percent of a Number

Example I) 65% of the human body by weight is oxygen. How many pounds of oxygen are in a 214 pound person?

$214(0.65) = \boxed{139.1 \text{ lbs oxygen}}$

Example II) 88% of apple juice is water. How many ounces of water are in a 16 ounce jug of apple juice?

$16(0.88) = \boxed{14.08 \text{ oz water}}$

Example III) 3% of the human body is nitrogen. How many pounds of nitrogen are in a 112 pound person?

$112(0.03) = \boxed{3.36 \text{ pounds nitrogen}}$

Example IV) 92% of watermelons are water. How much water is in a 6 pound watermelon?

$6(0.92) = \boxed{5.52 \text{ pounds water}}$

15) After 17.6 years, a radioactive material retains 63% of its original mass. What is the half life of this radioactive material?

I) Find K.

$$A(t) = A_0 e^{kt}$$

$$0.63 A_0 = A_0 e^{k(17.6)}$$

$$\frac{0.63 A_0}{A_0} = \frac{A_0 e^{17.6k}}{A_0}$$

$$0.63 = e^{17.6k}$$

$$17.6k = \log_e 0.63$$

$$17.6K = \ln 0.63$$

$$K = \frac{\ln 0.63}{17.6}$$

$$K = -0.02625\ldots$$

II) $A(t) = A_0 e^{kt}$

$$\frac{1}{2} A_0 = A_0 e^{(-0.02625\ldots)t}$$

$$\frac{\frac{1}{2} A_0}{A_0} = \frac{A_0 e^{(-0.02625\ldots)t}}{A_0}$$

$$\frac{1}{2} = e^{(-0.02625\ldots)t}$$

$$(-0.02625\ldots)t = \log_e\left(\frac{1}{2}\right)$$

$$(-0.02625\ldots)t = \ln\left(\frac{1}{2}\right)$$

$$t = \frac{\ln\left(\frac{1}{2}\right)}{-0.02625\ldots}$$

$$\boxed{t_{\frac{1}{2}} = 26.4 \text{ years}}$$

(16) After 23 days, a radioactive material retains 41% of its original mass. What is the half life of this radioactive material?

I) Find k.

$$A(t) = A_0 e^{kt}$$

$$0.41 A_0 = A_0 e^{k(23)}$$

$$\frac{0.41 A_0}{A_0} = \frac{A_0 e^{k(23)}}{A_0}$$

$$0.41 = e^{23k}$$

$$23k = \log_e 0.41$$

$$23k = \ln 0.41$$

$$k = \frac{\ln 0.41}{23}$$

$$k = -0.03876\ldots$$

II) $A(t) = A_0 e^{kt}$

$$\frac{1}{2} A_0 = A_0 e^{(-0.03876\ldots)t}$$

$$\frac{\frac{1}{2} A_0}{A_0} = \frac{A_0 e^{(-0.03876\ldots)t}}{A_0}$$

$$\frac{1}{2} = e^{(-0.03876\ldots)t}$$

$$(-0.07876\ldots)t = \log_e\left(\frac{1}{2}\right)$$

$$(-0.07876\ldots)t = \ln\left(\frac{1}{2}\right)$$

$$t = \frac{\ln\left(\frac{1}{2}\right)}{-0.03876\ldots}$$

$$\boxed{t_{\frac{1}{2}} = 17.88 \text{ days}}$$

The Logistic Growth Model

When environmental factors limit the growth of some quantity, often the quantity can be estimated with the following function:

$$P(t) = \frac{c}{1 + ae^{-bt}}$$

where $P(t)$ is the quantity, "t" is time passed, c is the maximum value of $P(t)$, also called the carrying capacity, and "a" and "b" are experimentally determined constants.

(17) The population of a strain of E. coli after t hours can be predicted with the equation below.

$$P(t) = \frac{752}{1 + 28e^{-0.4t}}$$

a) What is the carrying capacity of the strain?

$\boxed{752 \text{ E. coli}}$

b) What is the initial population?

$$P(0) = \frac{752}{1 + 28e^{-0.4(0)}} = \frac{752}{1 + 28e^{0}} = \frac{752}{1 + 28(1)} = \frac{752}{1 + 28} = \frac{752}{29}$$

$$= \boxed{26 \text{ E. coli}}$$

c) What will the population be after 1.3 hours?

$$P(1.3) = \frac{752}{1 + 28e^{-0.4(1.3)}} = \frac{752}{1 + 28e^{-0.52}} = \frac{752}{1 + 28(0.5945\ldots)} = \boxed{43 \text{ E. coli}}$$

(18) The population of a suspension of yeast after t hours can be predicted with the equation below.

$$P(t)=\frac{2449}{1+39e^{-0.31t}}$$

a) What is the carrying capacity of the population?

$\boxed{\text{2,449 organisms}}$

b) What is the initial population?

$$P(0)=\frac{2{,}449}{1+39e^{-0.31(0)}}=\boxed{\text{61 organisms}}$$

c) What will the population be after 10 hours?

$$P(10)=\frac{2{,}449}{1+39e^{-0.31(10)}}=\boxed{\text{888 organisms}}$$

(19) The population of a colony of bacteria after t hours can be predicted with the equation below.

$$P(t)=\frac{250{,}000}{1+24e^{-0.22t}}$$

a) What is the carrying capacity of the population?

$\boxed{\text{250,000 bacteria}}$

b) What is the initial population?

$$P(0)=\frac{250{,}000}{1+24e^{-0.22(0)}}=\boxed{\text{10,000 bacteria}}$$

c) What will the population be after 5 hours?

$$P(5)=\frac{250{,}000}{1+24e^{-0.22(5)}}=\boxed{\text{27,812 bacteria}}$$

(20) The population of a strain of E. coli after t hours can be predicted with the equation below.

$$P(t) = \frac{877}{1 + 29e^{-0.12t}}$$

a) What is the carrying capacity of the strain?

$\boxed{877 \text{ E. coli}}$

b) What is the initial population?

$$P(0) = \frac{877}{1+29e^{-0.12(0)}} = \boxed{29 \text{ E. coli}}$$

c) What will the population be after 22 hours?

$$P(22) = \frac{877}{1+29e^{-0.12(22)}} = \boxed{286 \text{ E. coli}}$$